Exercise 5: Geospatial Visualization

5 points + 2 bonus In total

Submission: Sunday, 24th May, 23:59h

In this exercise, you will develop an interactive data visualization dashboard using Plotly Dash to explore the agricultural data. You may **reuse** parts of former exercise sheets. You can use the functionalities of **pandas, numpy, scikit-learn**, and **plotly/dash** for the exercise, where not otherwise stated.

Task 1: Filtering Functionality

1 point In total

You must support filtering the data according to the **Year**, the **Item** and the attribute users want to explore in the map.

a. **Download** the dataset in OLAT. The dataset contains the following variables:
   - **Area**: area name
   - **Year**: year
   - **Item**: crop type
   - **Yield_hg_ha**: crop yield (hectograms per hectare)
   - **Rainfall_mm**: average annual rainfall (millimeters)
   - **Temp**: average annual temperature (°C)
   - **Pesticides_ton_ha**: total pesticide usage (tons per hectare)
   - **Continent**: Continent of the area

Please note that there are NaN values in the dataset that should be dropped.

b. Create a slider for users to select one **Year**.

c. Create a dropdown menu for users to select one **Item** only.

d. Create a dropdown menu for users to select one attribute of the 4 numerical ones (**Yield_hg_ha**, **Rainfall_mm**, **Temp**, and **Pesticides_ton_ha**) to explore on the map.

Task 2: Geospatial Map

2 points In total

Create a Choropleth map using Plotly Express to visualize the filtered subset obtained from Task 1:

a. Convert area names into ISO-3 codes using **pycountry**, and handle edge cases (e.g., Russia, Turkey) with a manual mapping.

b. Draw the map and color the areas based on the selected **Item** and attribute. Please use a legend to clearly explain the visual encoding. 1 point

c. Enable click interaction to select or deselect areas. Click on an area will select it and click again to deselect it. Allow **multiple de-/selections** through multiple clicks.

d. Dim the original map if there are specific areas selected. Add another highlight trace on top of the original map, so that selected areas will appear with full opacity and unselected areas remain dimmed. 1 point

Task 3: Comparison Scatter Plot

1point In total

Visualize the data of the selected areas in the geospatial map using a scatter plot. This helps users to focus on a smaller subset of interest.

a. The x-axis being the **Year**, and the y-axis the attribute selected in the dropdown menu in Task 1.

b. If no area is selected in the geospatial map, show the data of the selected attribute over all years of all areas.

c. If one or more areas are selected in the geospatial map, show the data of the selected attribute for only the selected areas across all years.

d. Color the points in the scatterplot according to **Area** and include a clear legend.

Task 4: Parallel Coordinates

1 point In total

In this task, you will utilize parallel coordinates to visualize the multivariate data for the selected subset.

a. Filter the data by the selected **Year** and **Item** according to Task 1.

b. Draw the parallel coordinates plot with 4 parallel axes, representing the 4 numerical attributes **Yield_hg_ha**, **Rainfall_mm**, **Temp**, and **Pesticides_ton_ha**. Add a clear legend for the axes.

c. Highlight all areas which are selected on the geospatial map in a different color.

Optional Task: Dimensionality Reduction Scatter Plot

In this task, you will apply dimensionality reduction to multivariate data and visualize the result to gain deeper insights into the dataset structure.

2 points In total

a. Filter the data by the selected **Year** and **Item** according to Task 1.

b. Apply t-SNE to the 4 numerical attributes **Yield_hg_ha**, **Rainfall_mm**, **Temp**, and **Pesticides_ton_ha** for the filtered subset. Use an implementation from scikit-learn to reduce the data to **two dimensions**.

1 bonus point

c. Create a **scatter plot** visualizing the two dimensions you get after dimensionality reduction. Use the two t-SNE components as axes. Color the points according to the **Continent** and include a clear legend.

d. Interaction:

- Link the t-SNE plot with the other 3 views (geospatial map, comparison scatter plot, and parallel coordinates plot). The selected areas in the geospatial map will be highlighted by higher opacity, and the unselected areas will be dimmed in the t-SNE plot too.

1 bonus Point

- Enable lasso selection in the t-SNE plot, so that when users select a group of areas, the selected areas will be highlighted in the geospatial map and the parallel coordinates plot, and will appear in the comparison scatter plot.
- The system will support multi-selection through two interactions: clicking in the geospatial map and lasso selection in t-SNE plot. Selection from both types should be combined and persist across all views.

  Note: the **ctx** function in Dash may be useful.

Hand-in

Please hand in your solution in OLAT as a **.zip** file until **Sunday, 24th May, 23:59h**. The .zip file should include the following:

- Your **python code** with the executable final application in a single file (.py) with a reasonable amount of comments explaining your implementation.

  Note: You are **not** allowed to use **other libraries** despite the ones in environment.yml.
- Include a few **screenshots** demonstrating your running application.
- In case there are additional remarks, include them as a .txt or .pdf file.
- Please hand-in your own unique solutions. We do not accept copied solutions or solutions created by or with ChatGPT!